

Problem Title:
The Greek Philosopher Problem

You are given the following facts:

1. All men are mortal.
2. All Greeks are men.
3. Socrates is a Greek.

You have to prove that :

Socrates is mortal

Using the Resolution Refutation method in First-Order Predicate Logic (FOL).

Instructions:

You must perform each transformation step of the reasoning process using Prolog. Create separate sections (or predicates) in Prolog that demonstrate every stage—from English sentences to automated resolution.

Show each resolution step and end with the empty clause ($\perp$) proving that the goal follows from the given facts.

Expected Prolog File Structure:

Your .pl file should clearly demonstrate each phase:

- % Step 1: Predicate logic representation
- % Step 2: Negation of goal
- % Step 3: Implication removal
- % Step 4: Negation movement
- % Step 5: Standardization
- % Step 6: Skolemization
- % Step 7: CNF clauses
- % Step 8: Resolution implementation and execution

Deliverables:

1. A single Prolog file (greek_philosopher.pl) containing:
 - The code and comments for all transformation steps.
 - The resolution refutation process showing the derivation of the empty clause.
2. A screenshot of the final proof output (showing that the empty clause or contradiction has been derived).